

ASSIGNMENT 2 — PYTHON FUNDAMENTALS TO FUNCTIONS

Course: Artificial Intelligence (Machine Learning & Deep Learning)

Name: _____

Roll No: _____

Date: _____

Instructions: Complete both sections below in a single Jupyter Notebook. Answer Section A using Markdown cells and complete Section B with working code cells. Every task must include a detailed Markdown note explaining the logic and approach — not just a one-line comment.

Section A — Short Theory Questions (answer in the same notebook using Markdown cells)

- 1 What is the difference between a variable and a keyword in Python?
- 2 Explain the difference between a list, a tuple, and a dictionary.
- 3 What is the difference between **break**, **continue**, and **pass**?
- 4 What is a lambda function? How is it different from a regular function?
- 5 Explain variable scope (local vs global) with a short example.

Section B — Practical Tasks (code cells in Jupyter Notebook)

- 1 Write a program that takes user input and performs type casting (e.g., string to int/float).
- 2 Create a list, tuple, dictionary, and set with sample data; demonstrate at least 3 operations on each (e.g., indexing, adding/removing items, membership check).
- 3 Write a program using if/elif/else to categorize a number (positive/negative/zero) and grade a student's marks (A/B/C/F).
- 4 Write a program using a **for** loop and list comprehension to generate a list of squares of numbers 1–10.
- 5 Write a program using a **while** loop to print the Fibonacci series up to n terms.
- 6 Write a function **is_prime(n)** that returns True/False, and use it in a loop to print all prime numbers between 1–50.
- 7 Write a function using default and keyword arguments (e.g., a function that greets a user with a customizable message).
- 8 Use **map()** and **filter()** with a lambda function to process a list (e.g., filter even numbers, then square them).

Submission Requirements: Submit a single **Jupyter Notebook (.ipynb)** file containing both sections. Section A answers and every Section B task must be preceded by a detailed **Markdown** cell explaining the logic, approach, and any observations. Code cells must show their output. Name your file **Assignment2_YourName.ipynb**.